Governors: Wisdom and Justice — Answers

Wisdom

Preceptor Table

What are the units, outcome, and covariates for your scenario's question?

- Units:

Life insurer: each row is one candidate for the US Senate in this year's elections. Only a third of the Senate's 100 seats is up for election in any cycle, so with two major-party candidates per race, that is roughly 70 people.

Researcher: each row is one candidate for any elected office — senate, governor, mayor, school board, et cetera — in a US election this year, so the units number in the tens of thousands.

- Outcomes:

Life insurer: `lived_after` — the number of years the candidate lives after election day. One observed (eventually) outcome per row.

Researcher: two *potential* outcomes per row — years lived after the election if the candidate wins, and years lived if the candidate loses.

- Covariates:

Life insurer: age at election, party, and election result — the question names all three — plus anything else predictive (sex, region). Note that for the insurer, the election result is just another covariate.

Researcher: none are forced by the question, though age at election and win margin will earn their way into the model.

- Treatment (if any):

Life insurer: none. This is a predictive model; nothing is manipulated.

Researcher: the election outcome — winning versus losing. This is the variable whose causal effect the question is about.

- Quantity of interest:

Life insurer: the expected years lived after the election for each candidate, given their covariates.

Researcher: the average difference between the two potential outcomes — the average causal effect of winning, across candidates for every kind of elected office, on years lived.

Describe the key components of a Preceptor Table for your scenario; Use words like "units," "outcomes," and "covariates."

Life insurer: The Preceptor Table should have one row per US Senate candidate this year. In this exercise, the columns are Candidate, Years Lived After Election, and the covariates Election Result, Age at Election, and Party.

Researcher: The Preceptor Table should have one row per candidate, for any elected office, in this year's elections. The columns are Candidate, the two potential outcomes (Years Lived if Win, Years Lived if Lose), and the Treatment (Election Outcome).

Preceptor Table --- Life insurer (predictive)[1]

Unit[2]	Outcome[3]	Covariates[4]		
Candidate	Years Lived After Election	Election Result	Age at Election	Party
Elizabeth Warren	21	Win	77	Democrat
Mitch McConnell	9	Win	84	Republican
...	...	...	...	...
Rick Scott	18	Lose	73	Republican

[1] If all the information in this table were available, we could answer the question: How many years, on average, will this year's US Senate candidates live after their election, given their age, party, and election result?

[2] Each row represents one candidate for the US Senate in the 2026 election cycle. Only a third of the Senate's 100 seats is up for election in any cycle, so with two major-party candidates per race, that is roughly 70 people.

[3] Years lived after election day. For candidates still alive, the cell holds the true (future) value, because the Preceptor Table shows the unobservable truth.

[4] Election Result (Win or Lose) is a covariate here, not a treatment — the insurer is not asking what winning does to anyone, only using it to forecast. Age at Election is in years. Party is as listed on the ballot.

Preceptor Table --- Researcher (causal)[1]			
Unit[2]	**Potential Outcomes**[3]		**Treatment**[4]
Candidate	Years Lived if Win	Years Lived if Lose	Election Outcome
Joe Smith	23	18	Won
David Jones	28	22	Lost
...	...	...	...
Susan Chen	19	17	Won

[1] If all the information in this table were available, we could answer the question: What is the average causal effect of winning an election, rather than losing it, on the number of years a candidate lives afterward?

[2] Each row represents one candidate for any elected office --- senate, governor, mayor, et cetera --- in a US election this year, so there are tens of thousands of such candidates.

[3] The two potential outcomes: years lived after the election if the candidate wins, and if the candidate loses. The cross-hatched cells mark each candidate's unobservable potential outcome — the one we could never see, because the candidate actually experienced the other election result.

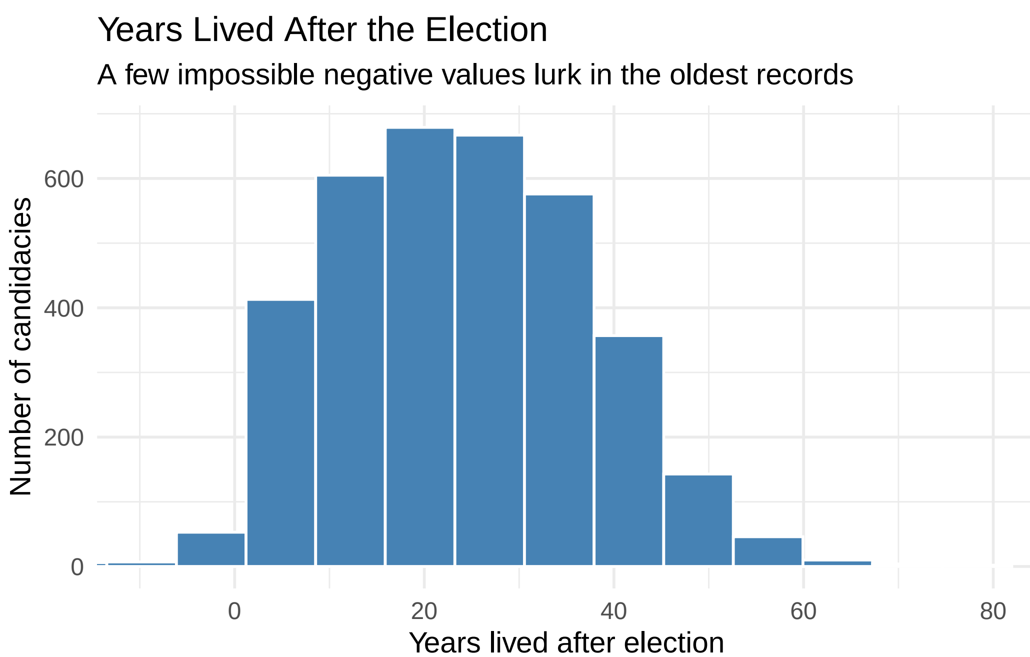
[4] The candidate's election outcome, taking values 'Won' or 'Lost'.

Exploratory Data Analysis

Create a plot that shows the distribution of `lived_after`.

```
governors |>
  ggplot(aes(x = lived_after)) +
  geom_histogram(bins = 40, fill = "steelblue", color = "white") +
  coord_cartesian(xlim = c(-10, 80)) +
  labs(
    title = "Years Lived After the Election",
    subtitle = "A few impossible negative values lurk in the oldest records",
    x = "Years lived after election",
    y = "Number of candidacies"
```

```
  ) +
  theme_minimal()
```



The outcome is continuous and widely spread, from candidates who died shortly after their election to some who lived another seven decades. A handful of *negative* values — candidates recorded as dying before their own election — flag data-quality problems in the oldest records, one more reason to restrict attention to the post-1945 era.

Create a tibble, called `x`, which keeps only close elections (`win_margin` within 5 points) after 1945 and adds a variable `election_result` recording whether the candidate won or lost.

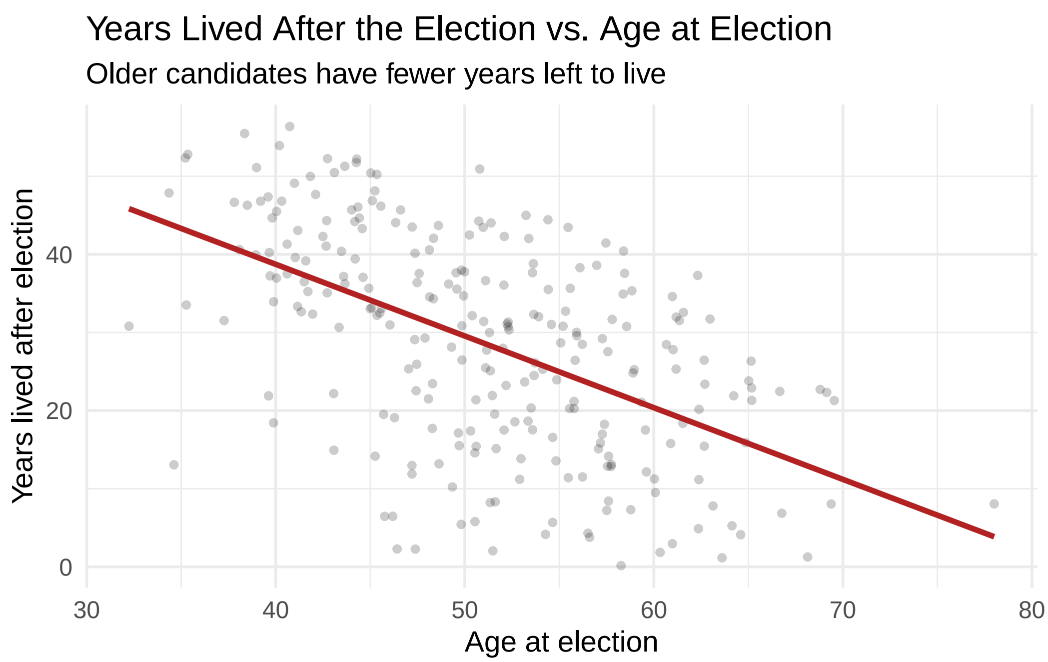
```
x <- governors |>
  filter(abs(win_margin) <= 5, year > 1945) |>
  mutate(election_result = if_else(win_margin > 0, "Win", "Lose"))
```

254 candidacies remain — the price of the close-election restriction is discarding 93% of the data. Reassuringly, winners and losers of these close races are almost exactly the same age at election (51.0 vs. 51.4), which is what we would expect if close elections really do assign winning as good as randomly.

Using `x`, create a plot that shows the relationship between `lived_after` and `election_age`.

```r
x |>
  ggplot(aes(x = election_age, y = lived_after)) +
  geom_point(alpha = 0.2, size = 1) +
  geom_smooth(method = "lm", se = FALSE, color = "firebrick") +
  labs(
    title = "Years Lived After the Election vs. Age at Election",
    subtitle = "Older candidates have fewer years left to live",
    x = "Age at election",
    y = "Years lived after election"
  ) +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



The strong negative relationship is close to mechanical: the older you are on election day, the fewer years you have left. Any model of `lived_after` that omits `election_age` will attribute this huge age effect to whatever covariates happen to correlate with age — so it must be in the model. (Plotted on the raw data instead, this scatter would also expose a few dozen impossible values — negative ages at election — hiding in the pre-1945 records; our filter removes every one of them.)

Justice

Population Table

Describe the key components of a Population Table for your scenario.

Life insurer: One row per candidate per election year, drawn from the population of candidates for major US statewide offices from roughly 1945 to 2030. The Data rows are gubernatorial candidacies from the Barfort et al. data; the Preceptor rows are this year's Senate candidates. The columns are Source, Candidate, Year, Years Lived After Election, Election Result, Age at Election, and Party.

Researcher: One row per candidate per election year, drawn from the same population but with the causal structure made explicit. The columns are Source, Candidate, Year, the two potential outcomes, and the Treatment.

Population Table --- Life insurer (predictive)[1]

	Unit/Time[2]		Outcome[3]	Covariates[4]		
Source	Candidate	Year	Years Lived After Election	Election Result	Age at Election	Party
...	...	...	...	...	...	...
Data	Earl Warren	1946	28	Win	55	Republican
Data	George Wallace	1962	36	Win	43	Democrat
Data	...	...	...	...	...	...
Data	Pat Brown	1966	30	Lose	61	Democrat
...	...	...	...	...	...	...
Preceptor	Elizabeth Warren	2026	21	Win	77	Democrat
Preceptor	Mitch McConnell	2026	9	Win	84	Republican
Preceptor	...	...	...	...	...	...
Preceptor	Rick Scott	2026	18	Lose	73	Republican
...	...	...	...	...	...	...

[1] This table combines the Barfort et al. (2021) gubernatorial data with the insurer's 2026 Senate candidates, both drawn from the population of candidates for major US statewide offices from roughly 1945 to 2030.

[2] Data rows are 3,587 gubernatorial candidacies from 1851 to 2012 (our analysis keeps the 254 close elections after 1945). Preceptor rows are candidates for the US Senate in 2026. The gap in office and in era is what the assumptions must bridge.

[3] Years between election day and death, computed from recorded dates. Where a candidate's exact birth or death date is unknown, the sources impute July 1 — so some values are approximations rather than measurements.

[4] Election Result is recorded from official vote totals. Age at Election is in years. Party is as recorded on the ballot; almost all candidates are Democrats or Republicans.

Population Table --- Researcher (causal)[1]					
	Unit/Time[2]		**Potential Outcomes**[3]		**Treatment**[4]
Source	Candidate	Year	Years Lived if Win	Years Lived if Lose	Election Outcome
...	...	...	...	...	...
Data	Earl Warren	1946	28	...	Won
Data	Pat Brown	1966	...	30	Lost
Data	...	...	...	...	...
Data	George Wallace	1962	36	...	Won
...	...	...	...	...	...
Preceptor	Joe Smith	2026	23	18	Won
Preceptor	David Jones	2026	28	22	Lost
Preceptor	...	...	...	...	...
Preceptor	Susan Chen	2026	19	17	Won
...	...	...	...	...	...

[1] This table combines the Barfort et al. (2021) gubernatorial data with the researcher's candidates in this year's elections, both drawn from the population of candidates for US elective office from roughly 1945 to 2030.

[2] Data rows are gubernatorial candidacies from 1946 to 2003 (close elections only in our analysis). Preceptor rows are candidates in 2026 races for any office. The gaps of decades, and of office, are the stability and representativeness concerns.

[3] In Data rows, only the potential outcome matching the actual election result was observed; the other is shown as '...' because no one measured it. In Preceptor rows, both potential outcomes are filled in with the truth, and the unobservable one is cross-hatched.

[4] In Data rows, the treatment is winning or losing an actual gubernatorial race; in close races, the result is nearly a coin flip, which is what makes as-if-random assignment credible. In Preceptor rows, it is winning or losing whatever race the candidate is running this year.

Assumptions

Why did we restrict our data to close elections?

> To make unconfoundedness plausible. A race decided by one point is close to a coin flip: which side of the threshold a candidate lands on is nearly unrelated to

their health, wealth, or anything else that also affects lifespan. This is the core of a *regression discontinuity design*: near the margin-of-victory cutoff, treatment assignment is as good as random. The price is steep — we discard 93% of the data — and it changes what we estimate: an effect *for close elections*, which may not describe landslides.

Counter-example to the assumption of validity:

> The treatment column in the data records winning a *gubernatorial* election. The insurer's table records the result of a *Senate* race, and the researcher's covers every elected office from senator to mayor. Those columns share a name, but winning a governorship (running a state) is a different experience — different stress, income, and post-office life — from winning a Senate seat. There is also an outcome-measurement crack: where exact birth or death dates were unknown, the sources impute July 1, so some `lived_after` values are approximations.

Counter-example to the assumption of stability:

> Most of our close elections come from the 1940s–1980s, when holding office carried different medical care, security, income, and stress than it does today. If winning office extended life in 1955 mainly through income and access to doctors — channels that matter less for wealthy modern candidates — the coefficient estimated on old data is not the coefficient for 2026.

Counter-example to the assumption of representativeness:

> Two leaps, both shaky. Data → population: close gubernatorial elections are not representative of all elections — candidates who end up in coin-flip races may differ systematically (more competitive states, weaker incumbents) from candidates generally. Population → Preceptor Table: Senate candidates (insurer) or candidates for any office (researcher) differ from gubernatorial candidates in age, wealth, and career paths.

Counter-example to the assumption of unconfoundedness:

> **Life insurer:** none needed. Unconfoundedness applies only to causal questions; the insurer's model is predictive.
>
> **Researcher:** wealthier and healthier candidates both win more often and live longer, so in the full data, winning is badly confounded — that is exactly why we restricted to close elections. Within close races, the residual worry is subtler: if the candidate who wins a squeaker differs from the one who loses it (better campaign funding, stronger party machine — both wealth proxies), the coin-flip story breaks down.

Probability Family and Link Function

What probability family should we use?

> Normal Distribution

Why should we use that probability family?

> The Normal distribution is the standard choice for continuous outcomes, and years lived after the election is continuous.

What is the link function for our outcome?

> The identity link. Our model will be a linear regression.
